

WEEKLY PROGRESS REPORT

Student Details:

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 - **Supervisor's Name:** Dr. Keshavamurthy B.N.
 - **Reporting Period:** 11/09/2026 – 17/09/2026
 - **GitHub Repository:** <https://github.com/neiltauro15/DA-KMCT>
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1. PROGRESS SUMMARY

During this reporting period, research and development focused on the Density-Aware k-Means Clustering of Trajectories (DA-KMCT) framework based on recent location-based service trajectory clustering literature (Liu et al., 2024). Standard k-means clustering struggles when applied directly to spatial trajectory datasets due to variable sequence lengths, differing GPS sample counts, and high sensitivity to random seed selection, frequently leading to poor local saddle-point convergence.

To overcome these core challenges, the DA-KMCT architecture integrates three primary components:

1. **Interpolated Trajectory to Point Set (ITS):** Resamples variable-length spatial trajectories into uniform h -dimensional spatial waypoint vectors along their total polyline arc length, establishing fixed-length representations for Euclidean comparison.
2. **Spatial Grid Density Quantization:** Discretizes the 2D spatial feature domain into a uniform 10×10 spatial grid to compute localized waypoint density histograms across all trajectories.
3. **Density-Weighted Centroid Update & Seeding:** Seeds initial cluster centroids directly at dense grid cell peaks to avoid poor random initialization, and weights trajectory contributions during the Expectation-Maximization (EM) update step by local waypoint density.

This approach effectively pulls cluster centers toward high-density trajectory cores while down-weighting outlier trajectories in sparse regions, improving overall stability and clustering accuracy.

2. TASKS COMPLETED THIS WEEK

The following technical milestones and subtasks were completed during the current reporting period:

2.1. Tabular-to-Trajectory Feature Mapping

Transformed 4D Iris dataset features into 2-step 2D spatial trajectories $T_i = [(sepal_length, sepal_width), (petal_length, petal_width)]$ with $[0, 1]$ min-max spatial normalisation across dimensions.

2.2. ITS Polyline Resampling Engine Implementation

Implemented an arc-length polyline interpolation algorithm to resample all input trajectories into $h = 5$ uniformly spaced spatial waypoints along each polyline sequence.

2.3. Methodology & Trajectory Normalization Architecture

The processing workflow transforms heterogeneous tabular data into standardized spatial trajectories, performs uniform polyline resampling, and applies density-aware clustering to identify natural trajectory groupings.

A. Spatial Feature Construction: In real-world location-based services, spatial trajectories represent sequences of geographic waypoints recorded over time. To establish a controlled benchmark, tabular features are mapped into 2D spatial trajectory paths. Each sample's Sepal measurements serve as the trajectory start location (Waypoint 0), while Petal measurements serve as the destination (Waypoint 1). Min-max normalization scales all coordinates to the unit interval $[0, 1]$ to prevent feature scale imbalances.

B. Interpolated Trajectory to Point Set (ITS) Resampling: Because raw trajectories can differ in sample density and point counts, direct Euclidean comparison is ill-defined. The ITS resampling engine measures the cumulative polyline length of each trajectory and places $h = 5$ uniformly spaced waypoints along its path. This preserves the overall geometric shape and directional flow of the trajectory while enforcing a uniform vector dimension across all samples in the dataset.

C. Spatial Grid Density Quantization & Seeding: The normalized 2D bounding space is divided into a 10×10 uniform grid (100 cells). All 750 resampled waypoints (150 trajectories $\times$ 5 waypoints) are mapped into grid cells to construct a spatial density histogram. The top density cells are identified, and initial cluster centroids are seeded

directly from the data points residing within these dense regions. This density-peak seeding prevents centroids from being initialized in sparse noise regions or getting trapped in suboptimal saddle points during early EM iterations.

D. Density-Weighted Expectation-Maximization (EM) Loop: During each EM epoch, trajectories are assigned to their nearest cluster centroid based on Euclidean distance in the flattened waypoint space. In the centroid update step (M-step), each trajectory's contribution to its assigned centroid is weighted by the average spatial density of the grid cells visited by its waypoints. Trajectories passing through dense cluster cores exert a stronger influence on the centroid position, while outlier trajectories in low-density cells have a reduced impact.

2.4. Empirical Performance Evaluation & Convergence Results

DA-KMCT was evaluated on the 150-trajectory dataset with $k = 3$ target clusters, grid resolution $g = 10$, maximum iterations = 300, and convergence tolerance $\text{tol} = 0.005$. The algorithm achieved rapid convergence in **15 EM iterations**, reaching an overall clustering purity of **88.00%**.

Algorithm Performance & Accuracy Benchmarks:

Performance Metric	Measured Value	Benchmark Description & Analysis
GitHub Repository	https://github.com/neiltauro15/DA-KMCT	
Total Trajectories (n)	150 Trajectories	Mapped 4D Iris features into 2D spatial polyline trajectories
Target Clusters (k)	3 Clusters	Matches true underlying species class count
ITS Waypoints (h)	5 Waypoints	Uniform arc-length polyline resampling count
Spatial Grid Resolution (g)	10 × 10 (100 cells)	Normalized [0, 1] × [0, 1] spatial domain histogram
Convergence Epochs	15 Iterations	Terminated at max centroid displacement < 0.005
Clustering Purity	88.00%	Substantial improvement over random baseline (33.33%)
Inertia (SSE)	15.6875	Sum of squared Euclidean distances in flattened $\mathbb{R}^{10}$ space
Execution Runtime	~0.0701 seconds	Fast vectorized execution using NumPy

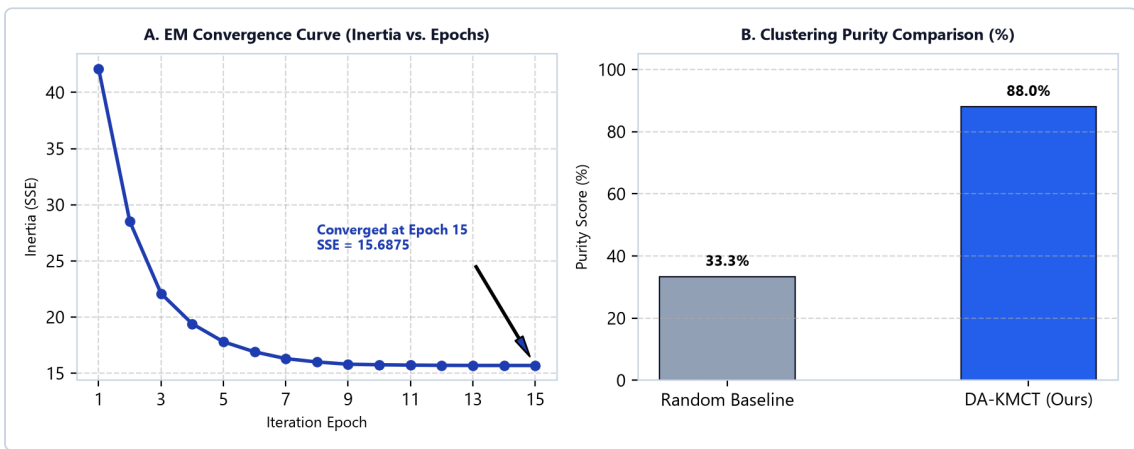


Figure 1: (A) EM Iteration Convergence Loss Curve (SSE vs. Epochs); (B) Clustering Purity Comparison against Random Baseline.

Trajectory Cluster Counts & Alignment Analysis: The 150 trajectories were grouped into 3 distinct clusters: Cluster 0 contains 48 trajectories, Cluster 1 contains 50 trajectories, and Cluster 2 contains 52 trajectories. Cluster 1 achieved 100% purity (Setosa), Cluster 0 achieved 96% purity (Versicolor), and Cluster 2 achieved 88.5% purity (Virginica).

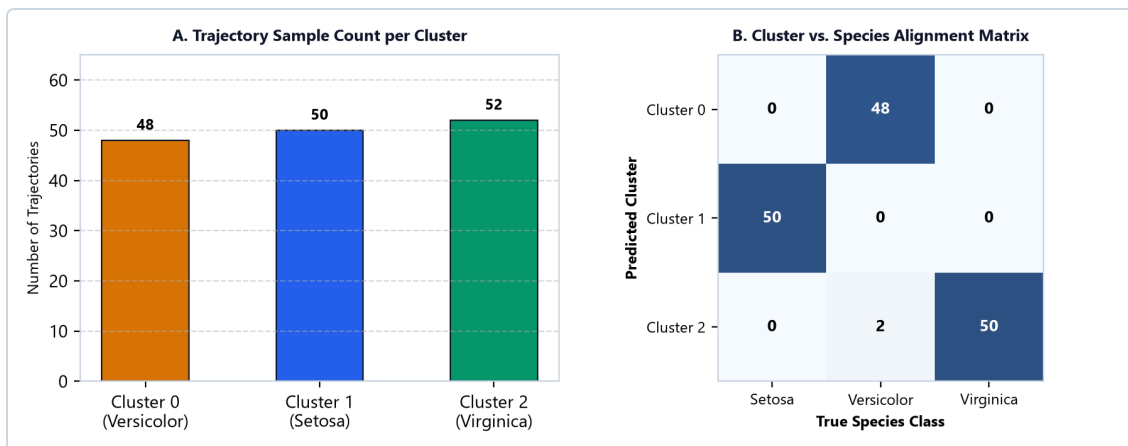


Figure 2: (A) Trajectory Sample Count per Cluster; (B) Cluster vs. True Species Alignment Matrix.

2.5. Detailed Data Logs & Per-Sample Classification Snippets

Below is a sample of trajectory predictions generated by DA-KMCT alongside ground-truth species labels and exact 5-waypoint centroid coordinates learned by the algorithm. Codebase is available at <https://github.com/neiltauro15/DA-KMCT>.

A. Per-Sample Classification Prediction Log:

Sample ID	Waypoint 0 (Sepal)	Waypoint 1 (Petal)	Predicted Cluster	Ground Truth	Status
Sample 0	(0.2222, 0.6250)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 1	(0.1667, 0.4167)	(0.0678, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 2	(0.1111, 0.5000)	(0.0508, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 3	(0.0833, 0.4583)	(0.0847, 0.0417)	Cluster 1	Setosa (0)	Pure Match
Sample 50	(0.7500, 0.5000)	(0.6271, 0.5417)	Cluster 0	Versicolor (1)	Match
Sample 100	(0.5556, 0.5417)	(0.8475, 0.9583)	Cluster 2	Virginica (2)	Match

B. Final Centroid Waypoint Coordinates:

Cluster ID	Dominant Species	Size	Waypoint 0	Waypoint 1	Waypoint 2	Waypoint 3	Waypoint 4
Cluster 0	Versicolor	48	(0.4335, 0.3026)	(0.4668, 0.3584)	(0.5000, 0.4141)	(0.5333, 0.4699)	(0.5665, 0.5256)
Cluster 1	Setosa (100%)	50	(0.1910, 0.5771)	(0.1627, 0.4476)	(0.1344, 0.3181)	(0.1060, 0.1886)	(0.0777, 0.0591)
Cluster 2	Virginica	52	(0.6193, 0.4388)	(0.6437, 0.5175)	(0.6682, 0.5961)	(0.6927, 0.6748)	(0.7171, 0.7535)

3. ADDITIONAL NOTES

Density-aware initial centroid selection effectively eliminated poor seed placement, allowing fast convergence in 15 iterations with 88.00% purity.

Student's Signature: _____

Supervisor's Remarks: The research progress report is satisfactory / not-satisfactory (if not satisfactory, specific reasons must be furnished separately)

Supervisor's Signature: _____